



# SAFETY DATA SHEET

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**Safety data sheet number**

M03393

**Product Name**

Total Nitrogen Test 'N Tube TM Reagent Set

**Product Code(s)**

2672245

**This product contains the following SDSs**

2671745VIAL  
2671846  
27242  
2671946  
2672046  
2672145VIAL

**Please scroll down to view the SDSs**



**Be Right™**

# SAFETY DATA SHEET

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**Product identifier**

**Product Name** TN (Total Nitrogen) Hydroxide Reagent  
**Safety data sheet number** M01349

**Other means of identification**

**Product Code(s)** 2671745VIAL  
**UN/ID no** 1760

**Manufacturer Address**

Hach Company  
P.O.Box 389 Loveland, CO 80539 USA  
(970) 669-3050

**Emergency Telephone**

(303) 623-5716 - 24 Hour Service (515)232-2533 - 8am - 4pm CST

**E-mail address** seasia@hach.com

**Product Information**

**Chemical Name** Not applicable  
**Formula** Not applicable  
**CAS No** Not applicable  
**Alternate CAS Number** Not applicable

**Substance**

Not applicable

**Mixture**

| Chemical Name                    | CAS No    | EC No     |
|----------------------------------|-----------|-----------|
| Demineralized Water<br>90 - 100% | 7732-18-5 | -         |
| Sodium Hydroxide<br>0 - 10%      | 1310-73-2 | 215-185-5 |
| Sodium Carbonate<br>0 - 10%      | 497-19-8  | 207-838-8 |

**GHS - Classification**

|                                   |            |
|-----------------------------------|------------|
| Skin corrosion/irritation         | Category 1 |
| Serious eye damage/eye irritation | Category 1 |

**Label elements**



**Signal word** - Danger

**Hazard statements**

H314 - Causes severe skin burns and eye damage

**Precautionary statements**

P260 - Do not breathe dust/fume/gas/mist/vapors/spray

P264 - Wash face, hands and any exposed skin thoroughly after handling

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P310 - Immediately call a POISON CENTER or doctor

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower

P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting

P405 - Store locked up

P501 - Dispose of contents/ container to an approved waste disposal plant

## 4. FIRST AID MEASURES

**Description of first aid measures**

**General advice**

Do not get in eyes, on skin, or on clothing. IF ON SKIN (or hair): Remove/Take off immediately all contaminated clothing. Rinse skin with water/shower. Immediately call a POISON CENTER or doctor/physician. Do not breathe dust/fume/gas/mist/vapors/spray.

**Eye contact**

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Keep eye wide open while rinsing. Call a physician immediately.

**Skin contact**

IF ON SKIN (or hair): Remove/Take off immediately all contaminated clothing. Rinse skin with water/shower. Wash contaminated clothing before reuse. Immediate medical attention is required. Wash hands thoroughly after handling.

**Inhalation**

IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing. Call a physician or poison control center immediately.

**Ingestion**

IF SWALLOWED. Do NOT induce vomiting. Rinse mouth. Drink plenty of water. Call a physician or poison control center immediately. Never give anything by mouth to an unconscious person.

**Self-protection of the first aider**

Use personal protective equipment as required. Avoid contact with skin, eyes or clothing.

**Most important symptoms and effects, both acute and delayed**

**Symptoms**

See Section 11: TOXICOLOGICAL INFORMATION.

**Indication of any immediate medical attention and special treatment needed**

**Note to physicians**

Treat symptomatically.

## 5. FIRE-FIGHTING MEASURES

### Flammable properties

Material is not classified as flammable according to GHS criteria.

### Suitable Extinguishing Media

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

### Unsuitable extinguishing media

No information available.

### Specific hazards arising from the chemical

The product causes burns of eyes, skin and mucous membranes. Thermal decomposition can lead to release of irritating and toxic gases and vapors. In the event of fire and/or explosion do not breathe fumes.

**Hazardous combustion products** Carbon oxides. sodium oxides.

### Protective equipment and precautions for firefighters

Wear self-contained breathing apparatus and protective suit.

## 6. ACCIDENTAL RELEASE MEASURES

### Personal precautions

Avoid contact with skin, eyes or clothing. Evacuate personnel to safe areas. Use personal protective equipment as required. Keep people away from and upwind of spill/leak.

### Environmental precautions

Should not be released into the environment. Prevent further leakage or spillage if safe to do so. Prevent product from entering drains. See Section 12 for additional ecological information.

### Methods for containment

Absorb spilled liquid with non-reactive sorbent material. Stop spilled material from being released to the environment.

### Methods for cleaning up

Avoid contact with spilled material. Avoid breathing spilled material. Soak up with inert absorbent material. Take up mechanically, placing in appropriate containers for disposal. Clean contaminated surface thoroughly. Prevent product from entering drains. Dispose of in accordance with local, state and federal regulations or laws. Use personal protective equipment as required.

## 7. HANDLING AND STORAGE

### Precautions for safe handling

#### Advice on safe handling

Avoid contact with skin, eyes or clothing. Remove contaminated clothing and shoes. Wash contaminated clothing before reuse. Do not breathe dust/fume/gas/mist/vapors/spray. Ensure adequate ventilation, especially in confined areas. In case of insufficient ventilation, wear suitable respiratory equipment. Wash thoroughly after handling. Use personal protective equipment as required.

### Conditions for safe storage, including any incompatibilities

#### Storage Conditions

Keep container tightly closed in a dry and well-ventilated place. Keep out of the reach of children. Keep in properly labeled containers. Store locked up.

#### Flammability class

Not applicable

#### Incompatible materials

May react violently in contact with: Strong acids.

## 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

### Exposure Guidelines

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| Chemical Name                                    | ACGIH TLV                   |                  | OSHA PEL                                   | Ontario TWA      |                | European Union |  |
|--------------------------------------------------|-----------------------------|------------------|--------------------------------------------|------------------|----------------|----------------|--|
| Sodium Hydroxide<br>(0 - 10%)<br>CAS#: 1310-73-2 | Ceiling: 2 mg/m³            |                  | TWA: 2 mg/m³<br>(vacated) Ceiling: 2 mg/m³ | Ceiling: 2 mg/m³ |                | -              |  |
| Chemical Name                                    | China                       | Japan            | Korea                                      | Australia        | Taiwan         | Philippines    |  |
| Sodium Hydroxide<br>(0 - 10%)<br>CAS#: 1310-73-2 | Ceiling: 2 mg/m³<br>Ceiling | Ceiling: 2 mg/m³ | Ceiling: 2 mg/m³                           | 2 mg/m³ Peak     | TWA: 2 mg/m³   | TWA: 2 mg/m³   |  |
| Chemical Name                                    |                             | Vietnam TWAs     |                                            |                  | Vietnam STELs  |                |  |
| Sodium Hydroxide<br>(0 - 10%)<br>CAS#: 1310-73-2 |                             | TWA - 0.5 mg/m³  |                                            |                  | STEL - 1 mg/m³ |                |  |

**Other Information** Vacated limits revoked by the Court of Appeals decision in AFL-CIO v. OSHA, 965 F.2d 962 (11th Cir., 1992).

**Engineering Controls** Eyewash stations. Showers. Ensure adequate ventilation, especially in confined areas.

**Personal Protective Equipment**

**Eyeface protection** Tight sealing safety goggles. Face protection shield. Avoid contact with eyes.

**Skin and body protection** Wear protective gloves and protective clothing. Wear impervious protective clothing, including boots, gloves, lab coat, apron or coveralls, as appropriate, to prevent skin contact.

**Respiratory protection** Ensure adequate ventilation, especially in confined areas. In case of inadequate ventilation wear respiratory protection.

**General Hygiene Considerations**

Avoid contact with skin, eyes or clothing. Take off all contaminated clothing and wash it before reuse. Contaminated work clothing should not be allowed out of the workplace. Wear suitable gloves and eye/face protection. When using do not eat, drink or smoke. Keep away from food, drink and animal feeding stuffs. Regular cleaning of equipment, work area and clothing is recommended. Wash face, hands and any exposed skin thoroughly after handling.

## 9. PHYSICAL AND CHEMICAL PROPERTIES

**Information on basic physical and chemical properties**

**Physical state** Liquid

**Gas Under Pressure** Not classified according to GHS criteria

**Appearance** aqueous solution **Color** colorless

**Odor** Odorless **Odor threshold** No data available

| <u>Property</u>                      | <u>Values</u>                                             | <u>Remarks • Method</u>                     |
|--------------------------------------|-----------------------------------------------------------|---------------------------------------------|
| <b>Molecular weight</b>              | No data available                                         |                                             |
| <b>pH</b>                            | 12.93                                                     |                                             |
| <b>Melting point/freezing point</b>  | ~ 0 °C / 32 °F                                            | Estimation based on theoretical calculation |
| <b>Boiling point / boiling range</b> | ~ 100 °C / 212 °F                                         | Estimation based on theoretical calculation |
| <b>Evaporation rate</b>              | 1 (water = 1) Estimation based on theoretical calculation |                                             |
| <b>Vapor pressure</b>                | 24.002 mm Hg / 3.2 kPa at 25 °C / 77 °F                   | Estimation based on theoretical             |

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calculation

**Vapor density (air = 1)** 0.62 (air = 1)

**Specific gravity (water = 1 / air = 1)** ~1

Estimation based on theoretical calculation

**Partition Coefficient (n-octanol/water)** Not applicable

**Soil Organic Carbon-Water Partition Coefficient** Not applicable  
**Autoignition temperature**

**Decomposition temperature** No data available

**Dynamic viscosity** ~ 1 cP (mPa s) at 20 °C / 68 °F

**Kinematic viscosity** No data available

#### Solubility(ies)

##### **Water solubility**

| <u>Water solubility classification</u> | <u>Water solubility</u> | <u>Water Solubility Temperature</u> |
|----------------------------------------|-------------------------|-------------------------------------|
| Soluble                                | > 1000 mg/L             | 25 °C / 77 °F                       |

##### **Solubility in other solvents**

| <u>Chemical Name</u> | <u>Solubility classification</u> | <u>Solubility</u> | <u>Solubility Temperature</u> |
|----------------------|----------------------------------|-------------------|-------------------------------|
| Acid                 | Soluble                          | > 1000 mg/L       | 25 °C / 77 °F                 |

#### Other Information

**Metal Corrosivity** Not classified as corrosive to metal according to GHS criteria

**Steel Corrosion Rate** No data available

**Aluminum Corrosion Rate** No data available

**Bulk density** Not applicable

**Explosive properties** Not classified according to GHS criteria.

**Explosion data** No data available

**Upper explosion limit** No data available

**Lower explosion limit** No data available

**Flammable properties** Material is not classified as flammable according to GHS criteria.

##### **Flammability Limit in Air**

**Upper flammability limit:** No data available

**Lower flammability limit:** No data available

**Flash point** No data available

**Method** No information available

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**Oxidizing properties**

Not classified according to GHS criteria.

**Reactivity properties**

Not classified as self-reactive, pyrophoric, self-heating or emitting flammable gases in contact with water according to GHS criteria.

## 10. STABILITY AND REACTIVITY

**Reactivity properties**

Not classified as self-reactive, pyrophoric, self-heating or emitting flammable gases in contact with water according to GHS criteria.

**Stability**

Stable under normal conditions.

**Special dangers of the product**

No information available.

**Conditions to avoid**

Heat. Evaporation. Exposure to air or moisture over prolonged periods.

**Incompatible materials**

May react violently in contact with: Strong acids.

**Hazardous Decomposition Products** Thermal decomposition can lead to release of irritating and toxic gases and vapors.

**Possibility of Hazardous Reactions** Hazardous polymerization does not occur

**Explosive properties**

Not classified according to GHS criteria.

**Upper explosion limit**

No data available

**Lower explosion limit**

No data available

**Autoignition temperature**

No data available

**Sensitivity to Static Discharge**

None reported

**Sensitivity to Mechanical Impact**

None reported

## 11. TOXICOLOGICAL INFORMATION

**Information on Likely Routes of Exposure**

|                                                    |                                                                                                  |
|----------------------------------------------------|--------------------------------------------------------------------------------------------------|
| <b>Product Information</b>                         | Corrosive to skin. Corrosive to eyes.                                                            |
| <b>Inhalation</b>                                  | Corrosive by inhalation.                                                                         |
| <b>Eye contact</b>                                 | Corrosive to eyes. Avoid contact with eyes. Wash thoroughly after handling.                      |
| <b>Skin contact</b>                                | Cause severe skin burns and eye damage. Avoid contact with skin. Wash thoroughly after handling. |
| <b>Ingestion</b>                                   | Causes burns.                                                                                    |
| <b>Aggravated Medical Conditions</b>               | Skin disorders. Eye disorders. Respiratory disorders.                                            |
| <b>Toxicologically synergistic products</b>        | None known.                                                                                      |
| <b>Toxicokinetics, metabolism and distribution</b> | No information available.                                                                        |

**Acute Toxicity Information**

**Acute toxicity**

Based on the classification principles the classification criteria are not met.

**STOT - single exposure**

Based on the classification principles the classification criteria are not met.

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**Aspiration hazard**

Based on the classification principles the classification criteria are not met.

**Product Acute Toxicity Data**

**Oral Exposure Route** No data available

**Dermal Exposure Route** No data available

**Inhalation (Dust/Mist) Exposure Route** No data available

**Inhalation (Vapor) Exposure Route** No data available

**Inhalation (Gas) Exposure Route** No data available

**Other Exposure Routes** No data available

**Unknown acute toxicity**

**Ingredient Acute Toxicity Data**

**Oral Exposure Route**

| Chemical Name                                    | Endpoint type              | Reported dose | Exposure time | Toxicological effects | Key literature references and sources for data                   |
|--------------------------------------------------|----------------------------|---------------|---------------|-----------------------|------------------------------------------------------------------|
| Sodium Carbonate<br>(0 - 10%)<br>CAS#: 497-19-8  | Rat<br>LD <sub>50</sub>    | 4090 mg/kg    | None reported | None reported         | IUCLID (The International Uniform Chemical Information Database) |
| Chemical Name                                    | Endpoint type              | Reported dose | Exposure time | Toxicological effects | Key literature references and sources for data                   |
| Sodium Hydroxide<br>(0 - 10%)<br>CAS#: 1310-73-2 | Rabbit<br>LD <sub>50</sub> | 500 mg/kg     | None reported | None reported         | No information available                                         |

**Dermal Exposure Route**

| Chemical Name                                    | Endpoint type              | Reported dose | Exposure time | Toxicological effects | Key literature references and sources for data                   |
|--------------------------------------------------|----------------------------|---------------|---------------|-----------------------|------------------------------------------------------------------|
| Sodium Carbonate<br>(0 - 10%)<br>CAS#: 497-19-8  | Mouse<br>LD <sub>50</sub>  | 2210 mg/kg    | None reported | None reported         | No information available                                         |
| Chemical Name                                    | Endpoint type              | Reported dose | Exposure time | Toxicological effects | Key literature references and sources for data                   |
| Sodium Hydroxide<br>(0 - 10%)<br>CAS#: 1310-73-2 | Rabbit<br>LD <sub>50</sub> | 1350 mg/kg    | None reported | None reported         | IUCLID (The International Uniform Chemical Information Database) |

**Inhalation (Dust/Mist) Exposure Route**

| Chemical Name                                   | Endpoint type           | Reported dose | Exposure time | Toxicological effects | Key literature references and sources for data                   |
|-------------------------------------------------|-------------------------|---------------|---------------|-----------------------|------------------------------------------------------------------|
| Sodium Carbonate<br>(0 - 10%)<br>CAS#: 497-19-8 | Rat<br>LC <sub>50</sub> | 1.15 mg/L     | 4 hours       | None reported         | IUCLID (The International Uniform Chemical Information Database) |

**Inhalation (Vapor) Exposure Route** No data available

**Inhalation (Gas) Exposure Route** No data available

**Other Exposure Routes** No data available

**Skin corrosion/irritation**

Causes severe burns.



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**Product Skin Corrosion/Irritation Data**

No data available.

**Ingredient Skin Corrosion/Irritation Data**

| Chemical Name                                    | Test method          | Species | Reported dose | Exposure time | Results            | Key literature references and sources for data                                |
|--------------------------------------------------|----------------------|---------|---------------|---------------|--------------------|-------------------------------------------------------------------------------|
| Sodium Hydroxide<br>(0 - 10%)<br>CAS#: 1310-73-2 | Patch test           | Human   | 20 mg         | 24 hours      | Corrosive to skin  | RTECS (Registry of Toxic Effects of Chemical Substances)                      |
| Sodium Carbonate<br>(0 - 10%)<br>CAS#: 497-19-8  | Standard Draize Test | Rabbit  | 500 mg        | 24 hours      | Mild skin irritant | ECHA (The European Chemicals Agency)<br>HSDB (Hazardous Substances Data Bank) |
| Chemical Name                                    | Test method          | Species | Reported dose | Exposure time | Results            | Key literature references and sources for data                                |
| Sodium Carbonate<br>(0 - 10%)<br>CAS#: 497-19-8  | Patch test           | Rabbit  | 500 mg        | 4 hours       | Mild skin irritant | ECHA (The European Chemicals Agency)                                          |

**Serious eye damage/eye irritation**

Corrosive to eyes.

**Product Serious Eye Damage/Eye Irritation Data**

No data available.

**Ingredient Eye Damage/Eye Irritation Data**

| Chemical Name                                    | Test method          | Species | Reported dose | Exposure time | Results           | Key literature references and sources for data           |
|--------------------------------------------------|----------------------|---------|---------------|---------------|-------------------|----------------------------------------------------------|
| Sodium Hydroxide<br>(0 - 10%)<br>CAS#: 1310-73-2 | Standard Draize Test | Rabbit  | 0.05 mg       | 24 hours      | Corrosive to eyes | RTECS (Registry of Toxic Effects of Chemical Substances) |
| Sodium Carbonate<br>(0 - 10%)<br>CAS#: 497-19-8  | Standard Draize Test | Rabbit  | 100 mg        | 24 hours      | Eye irritant      | HSDB (Hazardous Substances Data Bank)                    |
| Chemical Name                                    | Test method          | Species | Reported dose | Exposure time | Results           | Key literature references and sources for data           |
| Sodium Carbonate<br>(0 - 10%)<br>CAS#: 497-19-8  | Standard Draize Test | Rabbit  | 0.1 mL        | 4 seconds     | Eye irritant      | ECHA (The European Chemicals Agency)                     |

**Sensitization Information**

**Sensitization**

Based on the classification principles the classification criteria are not met.

**Product Sensitization Data**

**Skin Sensitization Exposure Route**

No data available.

**Respiratory Sensitization Exposure Route**

No data available.

**Ingredient Sensitization Data**

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**Skin Sensitization Exposure Route**

No data available.

**Respiratory Sensitization Exposure Route**

No data available.

**Chronic Toxicity Information**

**Chronic toxicity**

Chronic exposure to corrosive fumes/gases may cause erosion of the teeth followed by jaw necrosis. Bronchial irritation with chronic cough and frequent attacks of pneumonia are common. Gastrointestinal disturbances may also be seen. Avoid repeated exposure. Possible risk of irreversible effects.

**STOT - repeated exposure**

Based on the classification principles the classification criteria are not met.

**Product Repeat Dose Toxicity Data**

**Oral Exposure Route**

No data available.

**Dermal Exposure Route**

No data available.

**Inhalation (Dust/Mist) Exposure Route**

No data available.

**Inhalation (Vapor) Exposure Route**

No data available.

**Inhalation (Gas) Exposure Route**

No data available.

**Other Exposure Routes**

No data available.

**Ingredient Repeat Dose Toxicity Data**

**Oral Exposure Route**

No data available

**Dermal Exposure Route**

No data available

**Inhalation (Dust/Mist) Exposure Route**

No data available

**Inhalation (Vapor) Exposure Route**

No data available

**Inhalation (Gas) Exposure Route**

No data available

**Other Exposure Routes**

No data available

**Carcinogenicity**

Based on the classification principles the classification criteria are not met.

**Legend**

|                                                                                           |                |
|-------------------------------------------------------------------------------------------|----------------|
| <b>ACGIH (American Conference of Governmental Industrial Hygienists)</b>                  | Does not apply |
| <b>IARC (International Agency for Research on Cancer)</b>                                 | Does not apply |
| <b>NTP (National Toxicology Program)</b>                                                  | Does not apply |
| <b>OSHA (Occupational Safety and Health Administration of the US Department of Labor)</b> | X - Present    |

**Oral Exposure Route**

No data available

**Dermal Exposure Route**

No data available

**Inhalation (Dust/Mist) Exposure Route**

No data available

**Inhalation (Vapor) Exposure Route**

No data available

**Inhalation (Gas) Exposure Route**

No data available

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**Other Exposure Routes** No data available

**Ingredient Carcinogenicity Data**

**Oral Exposure Route** No data available

**Dermal Exposure Route** No data available

**Inhalation (Dust/Mist) Exposure Route** No data available

**Inhalation (Vapor) Exposure Route** No data available

**Inhalation (Gas) Exposure Route** No data available

**Other Exposure Routes** No data available

**Germ cell mutagenicity**

Based on the classification principles the classification criteria are not met.

**Product Germ Cell Mutagenicity *invitro* Data**

No data available.

**Ingredient Germ Cell Mutagenicity *invitro* Data** No data available

**Oral Exposure Route** No data available

**Dermal Exposure Route** No data available

**Inhalation (Dust/Mist) Exposure Route** No data available

**Inhalation (Vapor) Exposure Route** No data available

**Inhalation (Gas) Exposure Route** No data available

**Other Exposure Routes** No data available

**Ingredient Germ Cell Mutagenicity *invivo* Data**

**Oral Exposure Route** No data available

**Dermal Exposure Route** No data available

**Inhalation (Dust/Mist) Exposure Route** No data available

**Inhalation (Vapor) Exposure Route** No data available

**Inhalation (Gas) Exposure Route** No data available

**Other Exposure Routes** No data available

**Reproductive toxicity**

Based on the classification principles the classification criteria are not met.

**Oral Exposure Route** No data available

**Dermal Exposure Route** No data available

**Inhalation (Dust/Mist) Exposure Route** No data available

**Inhalation (Vapor) Exposure Route** No data available

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**Inhalation (Gas) Exposure Route**

No data available

**Other Exposure Routes**

No data available

**Ingredient Reproductive Toxicity Data**

**Oral Exposure Route**

No data available

**Dermal Exposure Route**

No data available

**Inhalation (Dust/Mist) Exposure Route**

No data available

**Inhalation (Vapor) Exposure Route**

No data available

**Inhalation (Gas) Exposure Route**

No data available

**Other Exposure Routes**

No data available

## 12. ECOLOGICAL INFORMATION

**Ecotoxicity**

Based on the classification principles, not classified as hazardous to the environment

**Product Ecological Data**

**Aquatic toxicity**

**Fish**

No data available

**Crustacea**

No data available

**Algae**

No data available

**Other Aquatic Species**

No data available

**Terrestrial toxicity**

**Soil**

No data available

**Vertebrates**

No data available

**Invertebrates**

No data available

**Ingredient Ecological Data**

**Aquatic toxicity**

**Fish**

| Chemical Name                                    | Exposure time | Species                    | Endpoint type    | Reported dose | Key literature references and sources for data                   |
|--------------------------------------------------|---------------|----------------------------|------------------|---------------|------------------------------------------------------------------|
| Sodium Hydroxide<br>(0 - 10%)<br>CAS#: 1310-73-2 | 96 hours      | <i>Oncorhynchus mykiss</i> | LC <sub>50</sub> | 45.4 mg/L     | IUCLID (The International Uniform Chemical Information Database) |

**Crustacea**

| Chemical Name                                    | Exposure time | Species            | Endpoint type    | Reported dose | Key literature references and sources for data                   |
|--------------------------------------------------|---------------|--------------------|------------------|---------------|------------------------------------------------------------------|
| Sodium Hydroxide<br>(0 - 10%)<br>CAS#: 1310-73-2 | 48 Hours      | <i>Daphnia sp.</i> | EC <sub>50</sub> | 40.4 mg/L     | IUCLID (The International Uniform Chemical Information Database) |

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**Algae** No data available

**Other Aquatic Species** No data available

**Terrestrial toxicity**

**Soil** No data available

**Vertebrates** No data available

**Invertebrates** No data available

**Other Information**

**Persistence and degradability**

None known.

**Product Biodegradability Data**

If available, see ingredient data below.

**Ingredient Biodegradability Data**

Test data reported below

| Chemical Name                                    | Test method   | Biodegradation | Exposure time | Results               |
|--------------------------------------------------|---------------|----------------|---------------|-----------------------|
| Sodium Hydroxide<br>(0 - 10%)<br>CAS#: 1310-73-2 | None reported | None reported  | None reported | Readily biodegradable |
| Sodium Carbonate<br>(0 - 10%)<br>CAS#: 497-19-8  | None reported | None reported  | None reported | Readily biodegradable |

**Bioaccumulation**

None known.

**Product Bioaccumulation Data**

Test data reported below. If available, see ingredient data below.

**Ingredient Bioaccumulation Data**

| Chemical Name                                   | Test method   | Exposure time | Species       | Bioconcentration factor (BCF) | Results                                      |
|-------------------------------------------------|---------------|---------------|---------------|-------------------------------|----------------------------------------------|
| Sodium Carbonate<br>(0 - 10%)<br>CAS#: 497-19-8 | None reported | None reported | None reported | None reported                 | Does not have the potential to bioaccumulate |

**Additional information**

**Product Information**

**Partition Coefficient (n-octanol/water)** Not applicable

**Ingredient Information**

**Mobility**

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Mobility in soil: High mobility. If available, see ingredient data below.

#### Product Information

**Soil Organic Carbon-Water Partition Coefficient** Not applicable

**Ingredient Information** No data available

#### Additional information

#### Water solubility

#### Product Information

| <u>Water solubility classification</u> | <u>Water solubility</u> | <u>Water Solubility Temperature</u> |
|----------------------------------------|-------------------------|-------------------------------------|
| Soluble                                | > 1000 mg/L             | 25 °C / 77 °F                       |

#### Ingredient Information

| <u>Chemical Name</u>                             | <u>Water solubility classification</u> | <u>Water solubility</u> | <u>Water solubility temperature °C</u> | <u>Water solubility temperature °F</u> |
|--------------------------------------------------|----------------------------------------|-------------------------|----------------------------------------|----------------------------------------|
| Sodium Hydroxide<br>(0 - 10%)<br>CAS#: 1310-73-2 | Completely soluble                     | 420000 mg/L             | 0 °C                                   | 32 °F                                  |
| Sodium Carbonate<br>(0 - 10%)<br>CAS#: 497-19-8  | Completely soluble                     | 220000 mg/L             | 20 °C                                  | 68 °F                                  |

#### Other adverse effects

No information available.

### 13. DISPOSAL CONSIDERATIONS

#### Waste treatment methods

**Disposal of wastes** Disposal should be in accordance with applicable regional, national and local laws and regulations.

**Contaminated packaging** Dispose of in accordance with federal, state and local regulations.

**US EPA Waste Number** D002

**Special instructions for disposal** Check with national, local municipal and state authorities and waste contractors for pertinent local information on the disposal of this article.

**Waste from residues/unused products** Disposal should be in accordance with applicable regional, national and local laws and regulations.

**Contaminated packaging** Dispose of in accordance with federal, state and local regulations.

### 14. TRANSPORT INFORMATION

#### IMDG

**Proper shipping name** Corrosive Liquid, N.O.S.  
**Hazard Class** 8

**Product Code(s)** 2671745VIAL  
**Issue Date** 01-Apr-2016  
**Version** 4

**Product Name** TN (Total Nitrogen) Hydroxide Reagent  
**Revision Date** 01-Apr-2016  
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**UN/ID no** 1760  
**Packing Group** II

**IATA**

**UN/ID no** 1760  
**Proper shipping name** Corrosive Liquid, N.O.S.  
**Hazard Class** 8  
**Packing Group** II

**DOT**

**Proper shipping name** Corrosive Liquid, N.O.S.  
**Hazard Class** 8  
**UN/ID no** 1760  
**Packing Group** II

**TDG**

**Proper shipping name** Corrosive Liquid, N.O.S.  
**Hazard Class** 8  
**UN/ID no** 1760  
**Packing Group** II

**ADR**

**Proper shipping name** Corrosive Liquid, N.O.S.  
**Hazard Class** 8  
**UN/ID no** 1760  
**Packing Group** II

## 15. REGULATORY INFORMATION

**International Inventories**

|                      |          |
|----------------------|----------|
| <b>TSCA</b>          | Complies |
| <b>DSL/NDSL</b>      | Complies |
| <b>EINECS/ELINCS</b> | Complies |
| <b>ENCS</b>          | Complies |
| <b>IECSC</b>         | Complies |
| <b>KECL</b>          | Complies |
| <b>PICCS</b>         | Complies |
| <b>AICS</b>          | Complies |

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**EINECS/ELINCS** - European Inventory of Existing Chemical Substances/European List of Notified Chemical Substances

**ENCS** - Japan Existing and New Chemical Substances

**IECSC** - China Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

| Chemical Name    | Dangerous Substances | organic solvents | Harmful Substances Whose Names Are to be Indicated on the Label | Class 2*       | Class 1*       | Poisonous and Deleterious Substances Control Law |
|------------------|----------------------|------------------|-----------------------------------------------------------------|----------------|----------------|--------------------------------------------------|
| Sodium Hydroxide | >1 %                 | Not applicable   | Not applicable                                                  | Not applicable | Not applicable | Deleterious -X                                   |

**Product Code(s)** 2671745VIAL  
**Issue Date** 01-Apr-2016  
**Version** 4

**Product Name** TN (Total Nitrogen) Hydroxide Reagent  
**Revision Date** 01-Apr-2016  
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| Chemical Name    | ISHA - Harmful Substances Prohibited for Manufacturing, Importing, Transferring, or Supplying | Korea. Harmful Substances Requiring Permission | Toxic Chemicals Control Law | Toxic Release Inventory Chemicals - Group 1 | Toxic Release Inventory Chemicals - Group 2 |
|------------------|-----------------------------------------------------------------------------------------------|------------------------------------------------|-----------------------------|---------------------------------------------|---------------------------------------------|
| Sodium Hydroxide | Not applicable                                                                                | Not applicable                                 | 97-1-136 *                  | Not applicable                              | >=1.0 %                                     |

| Chemical Name    | List of Dangerous Goods |
|------------------|-------------------------|
| Sodium Hydroxide | Present                 |

#### Vietnam Law on Chemicals

| Chemical Name                 | ANNEX I | ANNEX II | ANNEX III | ANNEX IV | ANNEX V | ANNEX VI | ANNEX VII |
|-------------------------------|---------|----------|-----------|----------|---------|----------|-----------|
| Sodium Hydroxide<br>1310-73-2 | -       | -        | -         | -        | X       | -        | X         |
| Sodium Carbonate<br>497-19-8  | -       | -        | -         | -        | -       | -        | X         |

### 16. OTHER INFORMATION

#### Key or legend to abbreviations and acronyms used in the safety data sheet

NIOSH IDLH Immediately Dangerous to Life or Health

NDF no data

TWA (time-weighted average)

STEL (Short Term Exposure Limit)

Ceiling - Maximum limit value

Vacated - These values have no official status. The only enforceable contaminant levels are those listed as OSHA Final PELs.

These lists are provided as reference only. Please note that some state regulations reference these "vacated" exposure limits in their state regulations.

**Prepared By** Hach Product Compliance Department

**Issue Date** 01-Apr-2016

**Revision Date** 01-Apr-2016

**Revision Note** New SDS.

#### Disclaimer

**USER RESPONSIBILITY:** Each user should read and understand this information and incorporate it in individual site safety programs in accordance with applicable hazard communication standards and regulations.

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**End of Safety Data Sheet**